Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/23/2018
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	The Role of Vacuole Membrane Fusion in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
PI Email:	mrojasp@ncsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919-513-8044
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City:	Raleigh	**State:**	NC
Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	07/01/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/1/2018 per NSSC information (Ed., 8/24/17) NOTE: Extended to 7/1/2017 per NSSC information (Ed., 4/5/16)		
Key Personnel Changes/Previous PI:	August 2018: No changes from last year.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The long-term goal of this research is to identify and characterize mechanisms of plant vacuole biogenesis that are critical for gravity perception. A role of the vacuole in controlling gravity perception is well accepted, but the molecular mechanisms by which the vacuole interacts with amyloplasts or regulates their sedimentation during gravitropism are yet to be identified. We recently uncovered a novel vacuole phenotype in agravitropic mutant alleles of VTI11 in Arabidopsis. We predict that this vacuole phenotype of vti11 mutants has negative effects on the organization of actin cables or the cortical endoplasmic reticulum, and therefore prevents amyloplast sedimentation. In addition, we have uncovered a novel regulation of vacuolar membrane fusion by phosphoinositides. Based on our results, we developed the hypothesis that mechanisms of vacuole remodeling that are mediated by phosphoinositide signaling are critical for the perception or transduction of the gravitropic signal in plants. A unique assay for vacuole fusion that we developed will be used to determine the role of specific cellular components in membrane fusion and gravity perception. The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. Overall, this research will improve our understanding of vacuole biogenesis, the interactions between the vacuole and other cellular components involved in gravity perception, and the role of phosphoinosidides in gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. In addition, the function of the vacuole in establishing the morphology and dynamics of other organelles will be characterized. These results will enhance the fundamental understanding of how plant cells organize into a defined set of independent organelles.
Task Progress:	This last year was focused on submission and multiple revisions of a major publication that is now available in PNAS (Brillada et al. 2018). Dr. Brillada developed new fluorescent protein fusions to HOPS subunits VPS41 and VPS33 that had better signal and complemented mutant phenotypes. This was achieved using a GoldenBraid approach. In addition, she improved protocols for VPS33 and VPS41 protein expression in bacteria and assayed their binding to liposomes in vitro. This allowed her to conclude that VPS33 and VPS41 binding to phosphoinositides is regulated by the curvature of the cellular membranes. The Principal Investigator and Dr. Brillada also generated new double marker lines for quantitative co-localization of these proteins with other endomembrane markers.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Articles in Peer-reviewed Journals	Brillada C, Rojas-Pierce M. "Vacuolar trafficking and biogenesis: a maturation in the field." Curr Opin Plant Biol. 2017 Dec;40:77-81. Epub 2017 Aug 31. https://doi.org/10.1016/j.pbi.2017.08.005 ; PubMed PMID: 28865974 , Dec-2017
Articles in Peer-reviewed Journals	Brillada C, Zheng J, Krüger F, Rovira-Diaz E, Christin Askani J, Schumacher K, Rojas-Pierce M. "Phosphoinositides control the localization of HOPS subunit VPS41, which together with VPS33 mediates vacuole fusion in plants." Proc Natl Acad Sci US A. 2018 Aug 28;115(35):E8305-E8314. https://doi.org/10.1073/pnas.1807763115 ; PMID: 30104351 ; PubMed Central PMCID: PMC6126739 , Aug-2018